Foundations of Computer Programming: Introduction to object-oriented programming

Learning objectives:

- Understand programming and its fundamental concepts.
- Understand different programming languages and recognise the difference
- Understand characteristics of High-level and Low-level Programming Languages
- Understand the Rules of a Language (Syntax & Semantics)
- Develop a Comprehensive Understanding of Object-Oriented Programming
- Recognise the distinction between Procedural and Object-Oriented Programming approaches.
- Understand the fundamental principles of object-oriented programming

Lesson exercise: Week 1

Answer the following questions individually. Once you have completed this task, you will collaborate with a group of 3-5 students to create a presentation on "Object-oriented programming, four OOP principles, and the history of Python".

1. What is Computer programming?
2. What are the fundamental concepts of computer programming?
3. What are the different types of programming languages?
4. What are the characteristics of high-level and low-level programming languages?
5. What is machine language?
6. What is assembly language?
7. What is object-oriented programming?
8. What are the different concepts in object-oriented programming and explain them?
9. What is the difference between procedural and object-oriented programming?
10. Why it is important to understand syntax and semantics?
11. What is the difference between compiled and interpreted programming languages?